

Math 8 — End of Q4 Test · ANSWER KEY

End-of-Q4 Content · 20 questions

Teacher key. Student answers may use equivalent formats (fractions, decimals, with or without units).

1. $v = \text{flambda}$. $f = 20$, $\text{lambda} = 3$. v ?

Answer: 60 m/s

2. $v = 340$, $f = 68$. lambda ?

Answer: 5 m

3. $v = 3 \times 10^8$, $\text{lambda} = 600 \text{ nm}$. f ?

Answer: 5×10^{14} or 5e14

4. $v = 40$, $\text{lambda} = 8$. f ?

Answer: 5 Hz

5. Light distance in 1 sec?

Answer: 3×10^8 or 3e8 or 300000000

6. Angle of incidence 45° . Reflection?

Answer: 45°

7. Concave lens image is ____?

Answer: virtual or Virtual or v

8. $r = 4 \text{ AU}$. T ?

Answer: 8 yr

9. $r = 16 \text{ AU}$. T ?

Answer: 64 yr

10. $T = 8 \text{ yr}$. r ?

Answer: 4 AU

11. Cylinder $r = 4$, $h = 6$.

Answer: 96pi or 96pi

12. Cone $r = 5$, $h = 6$.

Answer: 50pi or 50pi

13. Sphere $r = 4$.

Answer: 85.33pi or 85.3pi

14. Cylinder $r = 3$, $h = 8$.

Answer: 72π or 72π

15. Sphere $r = 5$.

Answer: 166.67π or 166.7π

16. Cyclist 15 mph for 3 hrs. Distance?

Answer: 45 mi

17. 20 m ladder, base 12 m. Height?

Answer: 16 m

18. Tank $8 \times 4 \times 3$ m. Volume?

Answer: 96 m^3

19. 20 V, 4 A. P?

Answer: 80 W

20. $\$18/\text{hr} \times 35$ hrs. Pay?

Answer: 630 or \$630 \$